Quick Work

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IMPLEMENT YOUR OWN PING PROGRAM.

Aim:

To implement a ping program using python raw sockets. to check the reachability of a host and calculate the round trip time.

Code:

```
import socket.
import struct
import time.
import os.
def checksum(data):
    total = 0
    for i in range (0, len(data-1, 2):
        total += (data[i] << 8) + data[i+1]
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0x ffff)
    total = total >> 16
    Return (~total) & 0xffff
def ping(host):
    try:
        packet_id = os. getpid() & 0xffff
        sequence = 1
        header = Struct. pack(
                "!BBHHH")
                8, 0, 0, packet_id, sequence)
```

```
data = b"PING"
chksum = Checksum (header + data)
header = Struct.pack("!BBHHH", 8, 0,
                     chksum, Packet_id, Sequence)
packet = header + data.
Start = time.time()
Sock.Sendto (Packet, dest-addr, 1))
Sock.Settimeout (2)
Sock.recv from (1024)
end = time.time()
rtt = (end - start) * 1000
Sock.close()
except Socket.timeout:
        print ("Request timed out")
except Permission Error:
    print ("Permission denied. Run The
    program as Administrator (root.")
host = input ("Enter destination host:")
ping (host)
```

INPUT / Output:

Enter destination host: 8.8.8.8

Reply From 8.8.8.8: time = 24.53 ms.

Result:

Thus, The ping program was successfully implemented in Python using raw Sockets.